

Nutrient Signaling to mTOR and Cell Growth

The mammalian target of rapamycin (mTOR) is a conserved protein kinase involved in a multitude of cellular processes including cell growth. Increased mTOR activation is observed in multiple human cancers and inhibition of mTOR has proven efficacious in numerous clinical trials. mTOR comprises two complexes, termed mTORC1 and mTORC2. Both complexes respond to growth factors, whereas only mTORC1 is controlled by nutrients, such as glucose and amino acids. Since the discovery of mTOR, extensive studies have intricately detailed the molecular mechanisms by which mTORC1 is regulated. Somewhat paradoxically, amino acid induced mTORC1 activation—arguably the most essential stimulus leading to mTORC1 activation—is the least understood. Here we review the current knowledge of nutrient dependent regulation of mTORC1.